

Conformal Contraction Mathematical Formulation

Given Values

Quantity	Value
Metric used for verification	$M = \text{diag}(1.0, 1.0)$
Lower metric bound	$m = 1.0$
Upper metric bound	$\bar{m} = 1.0$
Contraction rate	$\alpha = 0.8$
Young's inequality parameter	$\epsilon = 0.4$
Dynamic risk level	$\delta^{\text{dyn}} = 0.05$
Contraction risk level	$\delta^{\text{con}} = 0.05$
Dynamic conformal quantile	$q_{1-\delta}^{\text{dyn}} = 0.897275$
Contraction conformal quantile	$q_{1-\delta}^{\text{con}} = 2.898368$
Observed dynamic score	$S^{\text{dyn}} = 0.136886$
Observed contraction score	$S^{\text{con}} = 0.067456$
Initial energy	$\hat{E}(\gamma(\cdot, 0)) = 0.0$
Maximum sampled energy	$E_{\text{max}} = 0.018738$

Closed-Loop System

The real closed-loop system controlled by the learned policy is

$$\dot{x} = f(x, u_{\text{learned}}(x, E)).$$

Add and subtract the nominal model-based controller term:

$$\dot{x} = f(x, u_{\text{learned}}(x, E)) = \hat{f}(x, u_{\text{mpc}}(x, E)) + f(x, u_{\text{learned}}(x, E)) - \hat{f}(x, u_{\text{mpc}}(x, E)).$$

For this verification, the candidate path is the learned-planner path and the nominal path is the reference path.

Contraction Metric Condition

Assume a learned contraction metric $\hat{M}(x, t) \succ 0$ satisfying

$$mI \preceq \hat{M}(x, t) \preceq \bar{m}I$$

and approximately

$$\dot{\hat{M}} + \hat{M} \frac{\partial \hat{f}_{\text{cl}}}{\partial x} + \frac{\partial \hat{f}_{\text{cl}}}{\partial x} \hat{M} \preceq -2\alpha \hat{M}.$$

Plug in the implemented diagonal metric bounds.

$$1.0I \preceq \begin{bmatrix} 1.0 & 0 \\ 0 & 1.0 \end{bmatrix} \preceq 1.0I$$

Final result.

$$\boxed{\hat{M} = M = \text{diag}(1.0, 1.0) \text{ is positive definite and bounded.}}$$

Riemannian Energy

Define the Riemannian energy along the minimizing geodesic γ connecting x and x_n as

$$\hat{E}(\gamma(\cdot, t)) = \int_0^1 \delta\gamma(s, t)^\top \hat{M}(\gamma(s, t)) \delta\gamma(s, t) ds, \quad \delta\gamma(s, t) = \frac{\partial\gamma(s, t)}{\partial s}.$$

For sampled 2D waypoint verification with a constant diagonal metric, use

$$e_i = x_{c,i} - x_{n,i} = \begin{bmatrix} \Delta x_i \\ \Delta y_i \end{bmatrix}, \quad E_i = e_i^\top M e_i.$$

Plug in $M = \text{diag}(1.0, 1.0)$.

$$E_i = \begin{bmatrix} \Delta x_i & \Delta y_i \end{bmatrix} \begin{bmatrix} 1.0 & 0 \\ 0 & 1.0 \end{bmatrix} \begin{bmatrix} \Delta x_i \\ \Delta y_i \end{bmatrix} = \Delta x_i^2 + \Delta y_i^2$$

The maximum energy is computed over sampled candidate and nominal trajectory points:

$$E_{\max} = \max_i E_i = \max_i e_i^\top M e_i.$$

Plug in the computed verification value.

$$E_{\max} = 0.018738$$

Final equivalent sampled Euclidean radius.

$$r_{\max} = \sqrt{\frac{E_{\max}}{m}} = \sqrt{\frac{0.018738}{1.0}} = 0.136886 \text{ m}$$

$$\boxed{E_{\max} \text{ is computed, not assumed.}}$$

Dynamic Conformal Score

Conformal inference gives the dynamic model-error event

$$\mathbb{P} \left[\left\| f(x, u_{\text{learned}}(x, E)) - \hat{f}(x, u_{\text{mpc}}(x, E)) \right\| \leq q_{1-\delta}^{\text{dyn}}, \forall t \in [0, T] \right] \geq 1 - \delta^{\text{dyn}},$$

where

$$q_{1-\delta}^{\text{dyn}} = \left\lceil (1 - \delta^{\text{dyn}})(N + 1) \right\rceil\text{-th smallest of } (S_1^{\text{dyn}}, \dots, S_N^{\text{dyn}})$$

and

$$S_i^{\text{dyn}} = \sup_{t \in [0, T]} \left\| f(x, u_{\text{learned}}(x_i, E_i)) - \hat{f}(x, u_{\text{mpc}}(x_i, E_i)) \right\|.$$

For sampled paths, compute

$$S^{\text{dyn}} = \max_i \|v_{c,i} - v_{n,i}\|_2.$$

Plug in the values.

$$S^{\text{dyn}} = 0.136886, \quad q_{1-\delta}^{\text{dyn}} = 0.897275$$

$$0.136886 \leq 0.897275$$

Final result.

$$\boxed{S^{\text{dyn}} \leq q_{1-\delta}^{\text{dyn}}}$$

Contraction Residual Score

The conformal contraction residual is

$$S_i^{\text{con}} = \max \left\{ 0, \sup_{t \in [0, T]} \left(2\delta \gamma(1, t)^\top \hat{M}(\gamma(1, t)) f(x, u_{\text{learned}}(x, E)) \right. \right. \\ \left. \left. - 2\delta \gamma(0, t)^\top \hat{M}(\gamma(0, t)) \hat{f}(x_n, u_{\text{mpc}}(x_n, E_n)) \right. \right. \\ \left. \left. + 2\alpha \hat{E}(\gamma(\cdot, t)) \right) \right\}.$$

For sampled paths with constant metric, compute

$$\rho_i = 2e_i^\top M v_{c,i} - 2e_i^\top M v_{n,i} + 2\alpha e_i^\top M e_i, \quad S^{\text{con}} = \max_i \max(0, \rho_i).$$

Plug in the values.

$$S^{\text{con}} = 0.067456, \quad q_{1-\delta}^{\text{con}} = 2.898368$$

$$0.067456 \leq 2.898368$$

Final result.

$$\boxed{S^{\text{con}} \leq q_{1-\delta}^{\text{con}}}$$

Conformal Quantiles

The conformal quantile rule is

$$q_{1-\delta} = \lceil (1 - \delta)(N + 1) \rceil\text{-th smallest calibration score.}$$

For this verification run, $N = 19$. Plug in the dynamic risk value.

$$k_{\text{dyn}} = \lceil (1 - 0.05)(19 + 1) \rceil = \lceil 19 \rceil = 19$$

$$q_{1-\delta}^{\text{dyn}} = 0.897275$$

Plug in the contraction risk value.

$$k_{\text{con}} = \lceil (1 - 0.05)(19 + 1) \rceil = 19$$

$$q_{1-\delta}^{\text{con}} = 2.898368$$

Final result.

$$\boxed{q_{1-\delta}^{\text{dyn}} = 0.897275, \quad q_{1-\delta}^{\text{con}} = 2.898368}$$

Union-Bound Coverage

Combine the dynamic and contraction conformal events using Boole's inequality:

$$\mathbb{P}[\text{both conformal events}] \geq 1 - \delta^{\text{dyn}} - \delta^{\text{con}}.$$

Plug in the values.

$$1 - 0.05 - 0.05 = 0.90$$

Final result.

$$\boxed{\mathbb{P}[\text{both conformal events}] \geq 0.90}$$

Young's Inequality Step

Use Young's inequality:

$$2q_{1-\delta}^{\text{dyn}}\sqrt{\bar{m}}\hat{d}(x, x_n) \leq \epsilon\hat{E}(\gamma(\cdot, t)) + \left(q_{1-\delta}^{\text{dyn}}\right)^2 \bar{m}\epsilon^{-1}, \quad \forall \epsilon > 0.$$

This yields

$$\mathbb{P}\left[\dot{\hat{E}}(\gamma(\cdot, t)) \leq -(2\alpha - \epsilon)\hat{E}(\gamma(\cdot, t)) + \left(q_{1-\delta}^{\text{dyn}}\right)^2 \bar{m}\epsilon^{-1} + q_{1-\delta}^{\text{con}}, \quad \forall t \in [0, T]\right] \geq 1 - \delta^{\text{dyn}} - \delta^{\text{con}}.$$

Plug in the values in the additive disturbance term.

$$\begin{aligned} \left(q_{1-\delta}^{\text{dyn}}\right)^2 \bar{m}\epsilon^{-1} + q_{1-\delta}^{\text{con}} &= (0.897275)^2(1.0)(0.4)^{-1} + 2.898368 \\ &= 2.012756 + 2.898368 = 4.911124 \end{aligned}$$

Final result.

$$\boxed{\dot{\hat{E}} \leq -1.2\hat{E} + 4.911124}$$

Effective Contraction Rate

Select ϵ so that

$$2\bar{\alpha} = 2\alpha - \epsilon > 0.$$

Plug in the values.

$$\begin{aligned} 2\bar{\alpha} &= 2(0.8) - 0.4 = 1.2 \\ \bar{\alpha} &= 0.6 \end{aligned}$$

Final result.

$$\boxed{\bar{\alpha} = 0.6 > 0}$$

Final Energy Bound

The final conformal contraction energy bound is

$$\mathbb{P} \left[\hat{E}(\gamma(\cdot, t)) \leq e^{-2\bar{\alpha}t} \hat{E}(\gamma(\cdot, 0)) + \frac{\left(q_{1-\delta}^{\text{dyn}}\right)^2 \bar{m}\epsilon^{-1} + q_{1-\delta}^{\text{con}}}{2\bar{\alpha}} (1 - e^{-2\bar{\alpha}t}), \forall t \in [0, T] \right] \geq 1 - \delta^{\text{dyn}} - \delta^{\text{con}}.$$

Plug in the values.

$$\hat{E}(\gamma(\cdot, t)) \leq e^{-2(0.6)t} (0.0) + \frac{(0.897275)^2 (1.0)(0.4)^{-1} + 2.898368}{2(0.6)} (1 - e^{-2(0.6)t})$$

$$\hat{E}(\gamma(\cdot, t)) \leq 4.092603 (1 - e^{-1.2t})$$

Final result.

$$\boxed{\hat{E}(\gamma(\cdot, t)) \leq 4.092603 (1 - e^{-1.2t})}$$

Tube Radius

Convert the energy bound to a Euclidean tube radius using the lower metric bound:

$$r(t) = \sqrt{\frac{\hat{E}(\gamma(\cdot, t))}{m}}.$$

Plug in the values.

$$r(t) \leq \sqrt{\frac{4.092603(1 - e^{-1.2t})}{1.0}}$$

Final result.

$$\boxed{r(t) \leq \sqrt{4.092603(1 - e^{-1.2t})} \text{ m}}$$

At steady state:

$$r_{\infty} = \sqrt{\frac{4.092603}{1.0}} = 2.023018 \text{ m}$$

$$\boxed{r_{\infty} = 2.023018 \text{ m}}$$

Final Verification Decision

The verification accepts the sampled path because both conformal scores are below their quantiles:

$$S^{\text{dyn}} \leq q_{1-\delta}^{\text{dyn}}, \quad S^{\text{con}} \leq q_{1-\delta}^{\text{con}}.$$

Plug in the values.

$$0.136886 \leq 0.897275, \quad 0.067456 \leq 2.898368$$

Final result.

$$\boxed{\text{Accepted conformal contraction verification with probability lower bound 0.90.}}$$